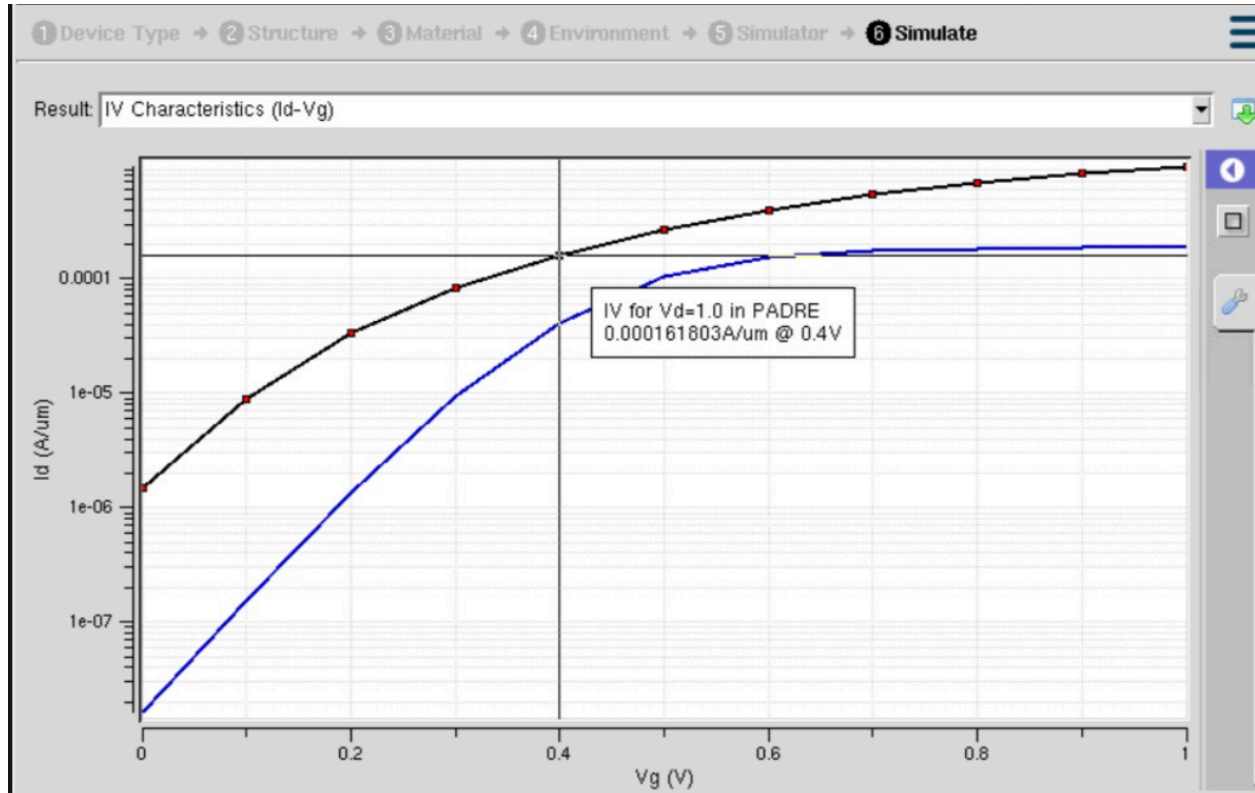


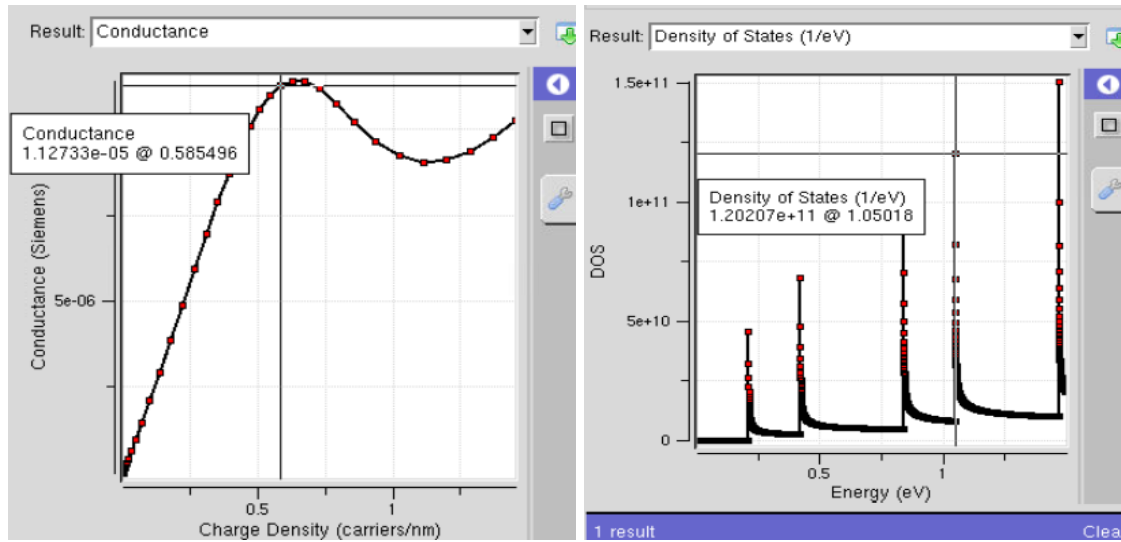
Exp. 8. Simulation of VI Characteristics of FINFET, CNTFET and GFET using NanoHub

Drain current Performance Vs Gate supply voltage of FinFET device

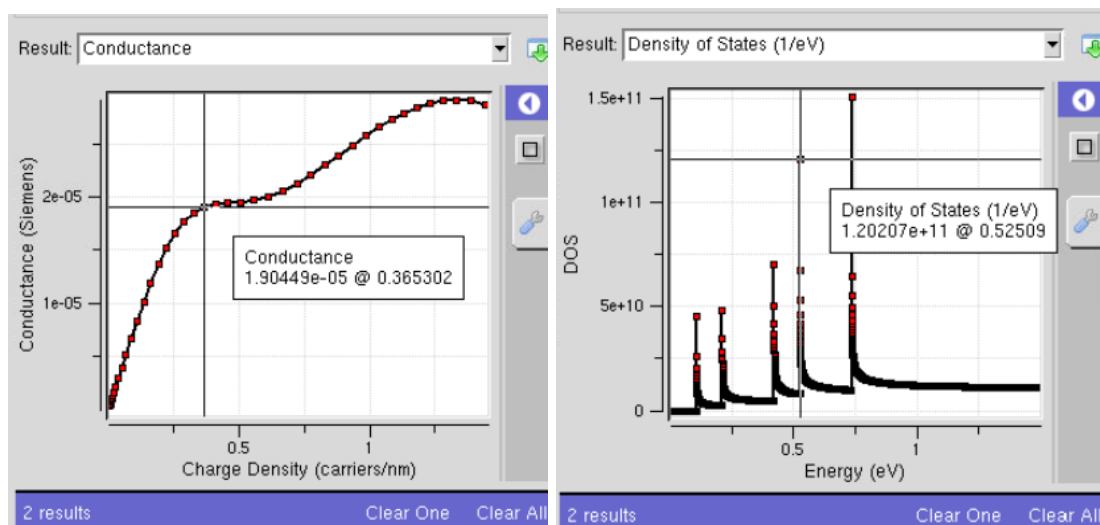


B) CNTFET

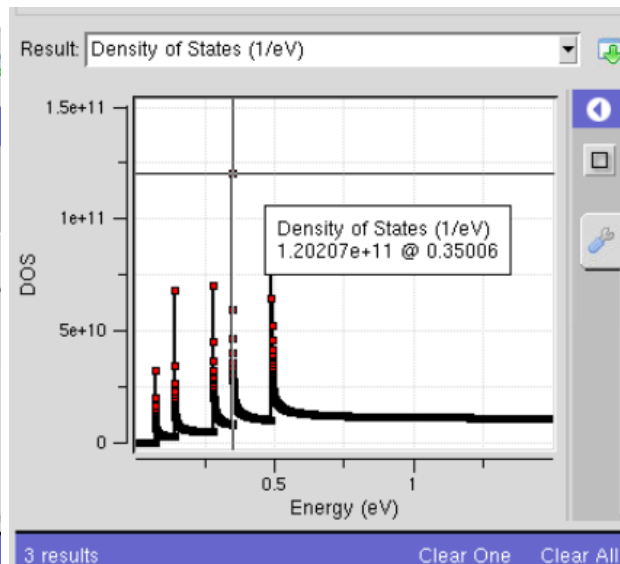
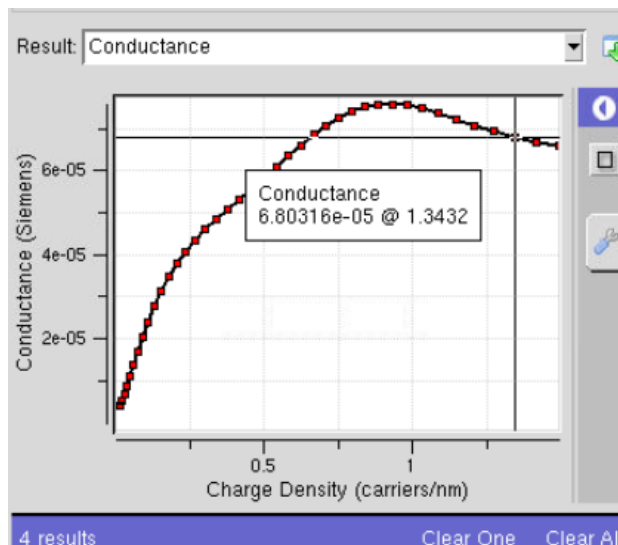
For Diameter (2nm) and Length(4nm)



For Diameter (4nm) and Length(4nm)

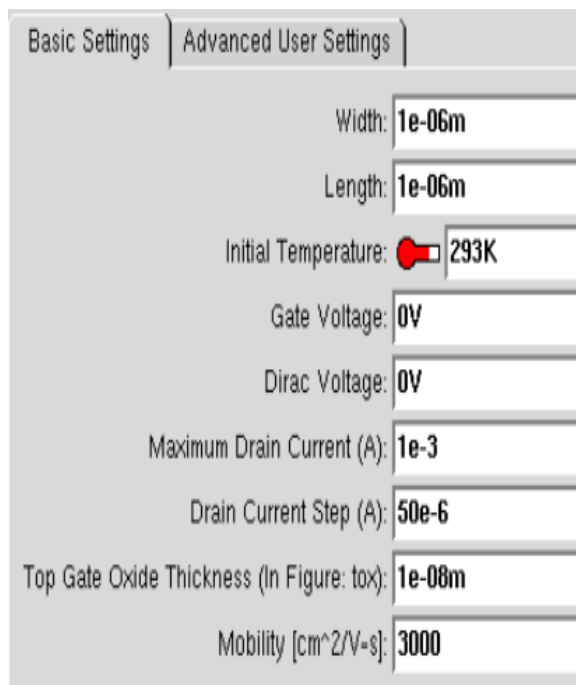
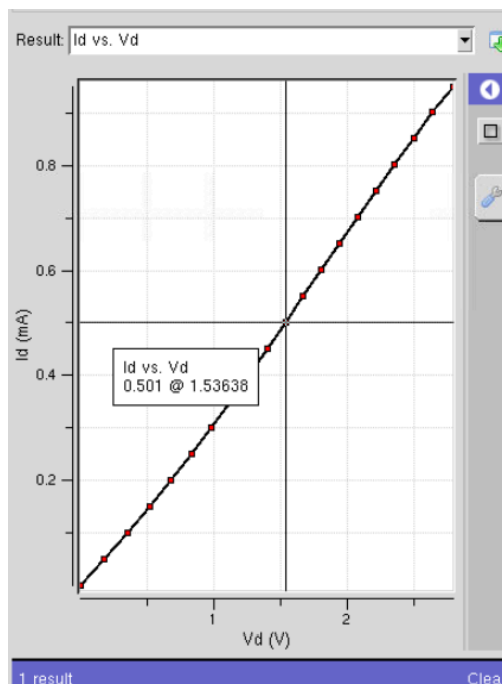


For Diameter(6nm) and Length(2nm)

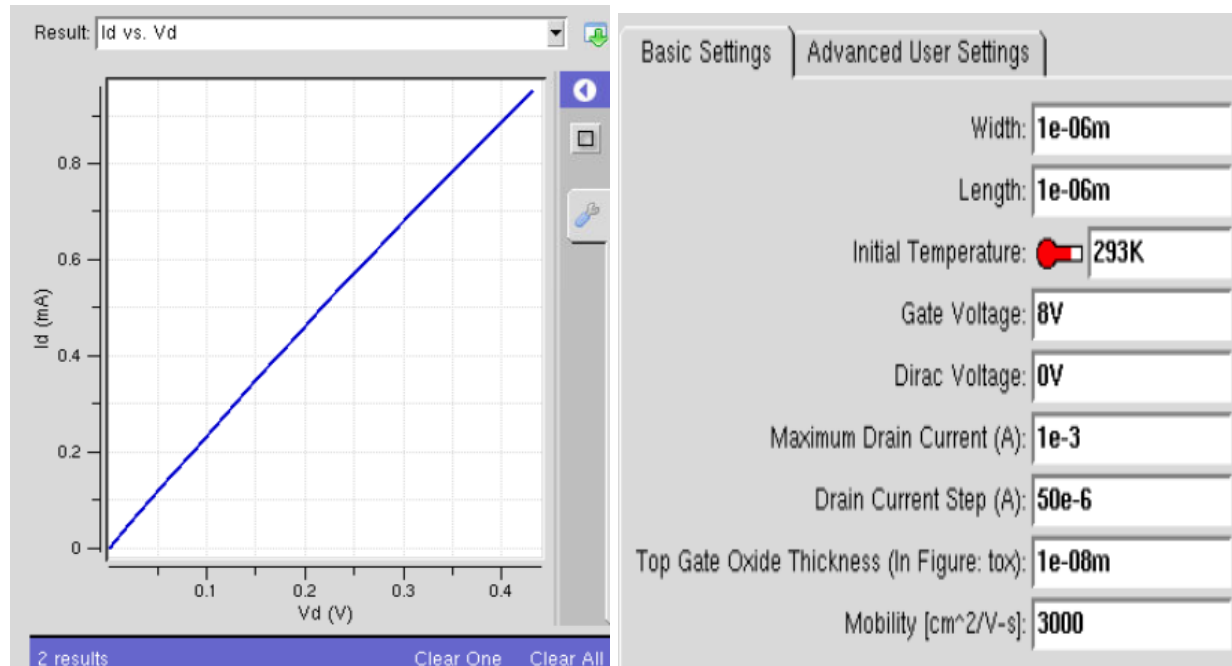


Conclusion: The electrical characteristics of CNTFET depended strongly on carbon nanotube chirality. Semiconducting nanotubes exhibited better switching performance than metallic nanotubes. The simulation confirms the importance of nanotube structure.

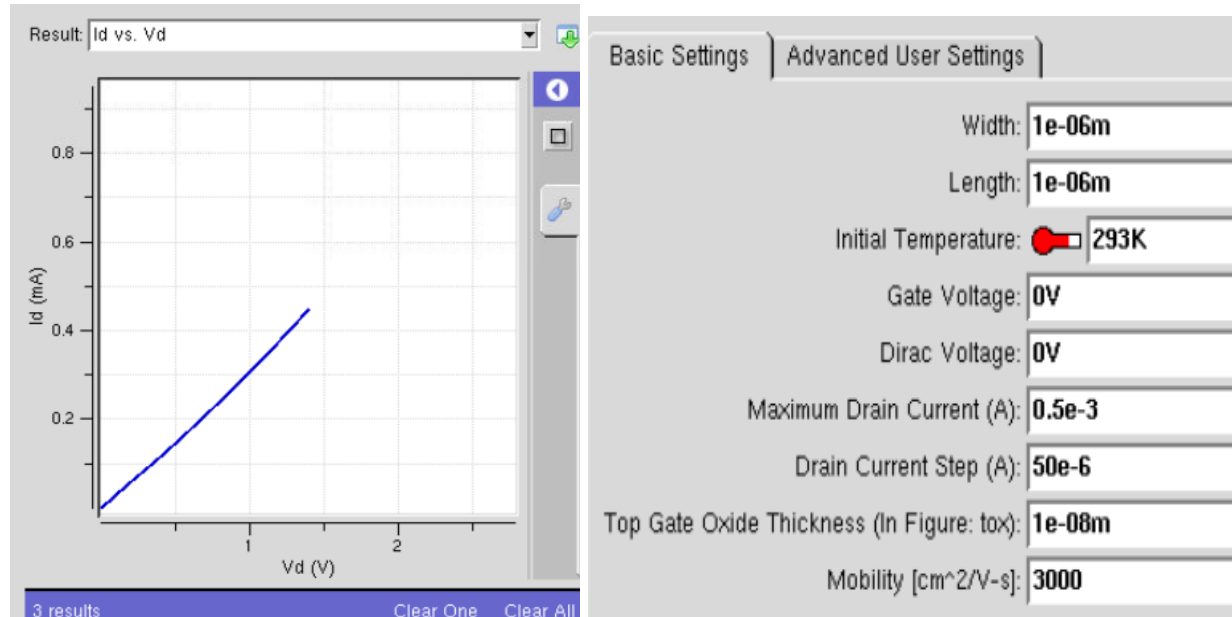
C)GFET



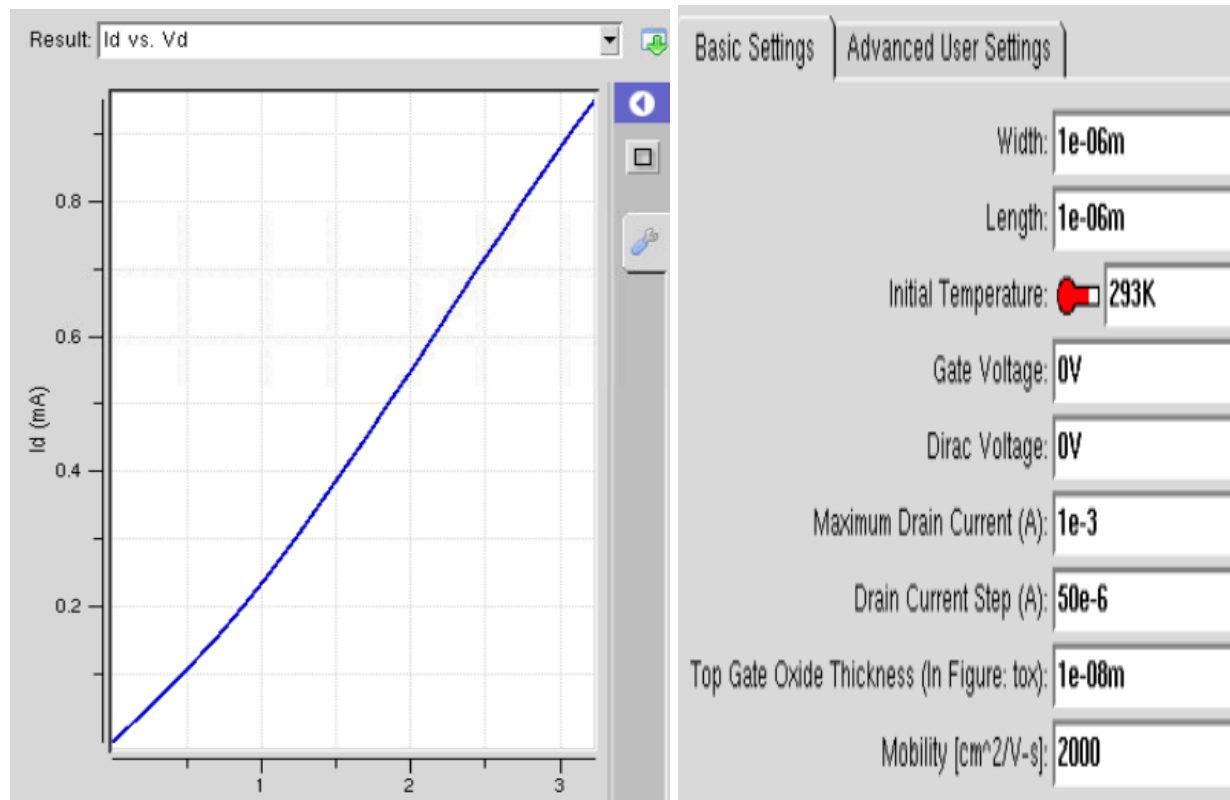
Case Study 1: How does varying the gate voltage (V_{gs}) affect the drain current (I_{ds}) in a GFET?



Case Study 2: What factors determine the maximum drain current ($I_{ds,max}$) in a Graphene Field Effect Transistor (GFET)?



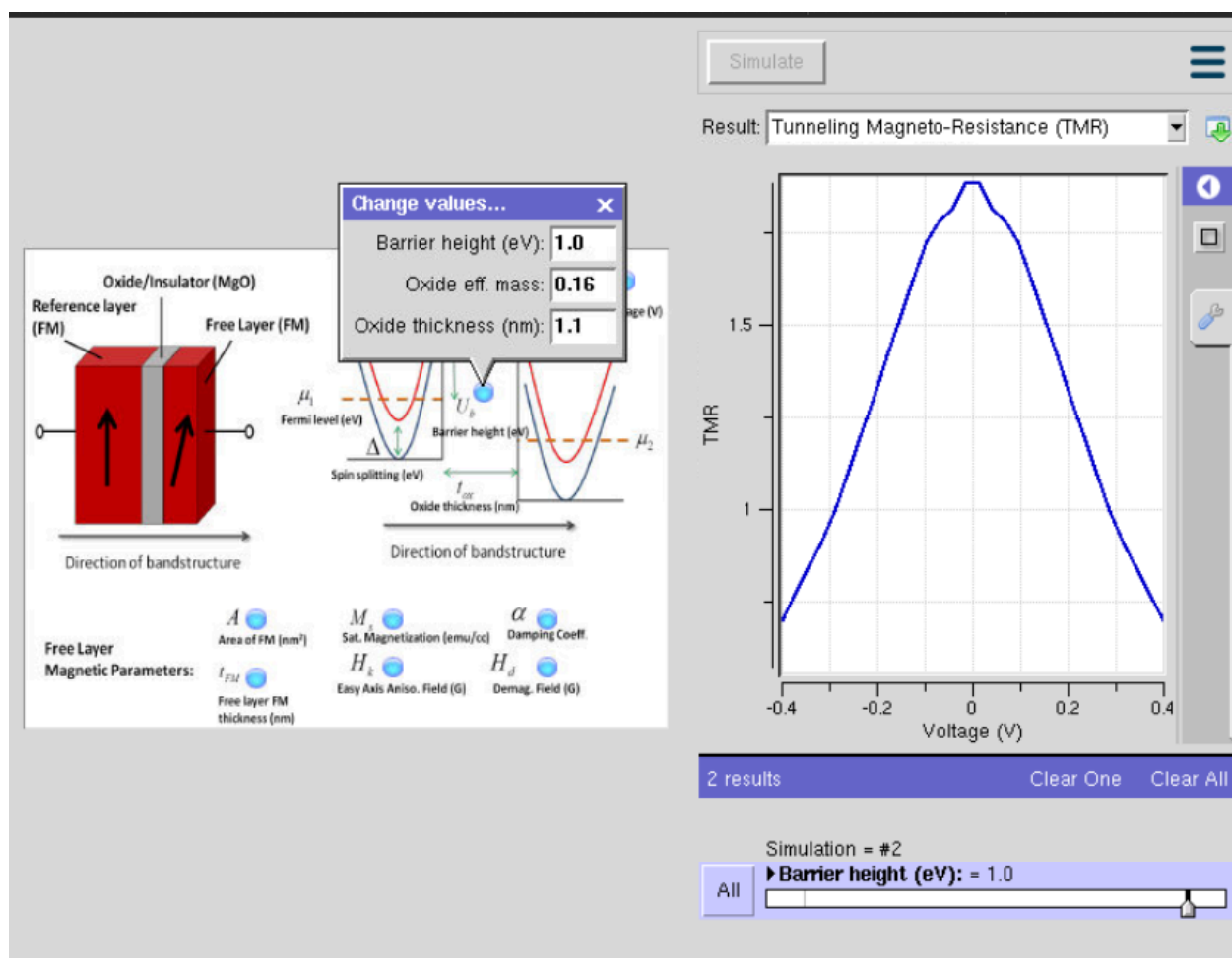
Case Study 3: To investigate how the mobility of charge carriers in a Graphene FieldEffect Transistor (GFET) with a mobility of $2000 \text{ cm}^2/\text{V}\cdot\text{s}$ influences its current-voltage (VI) characteristics.



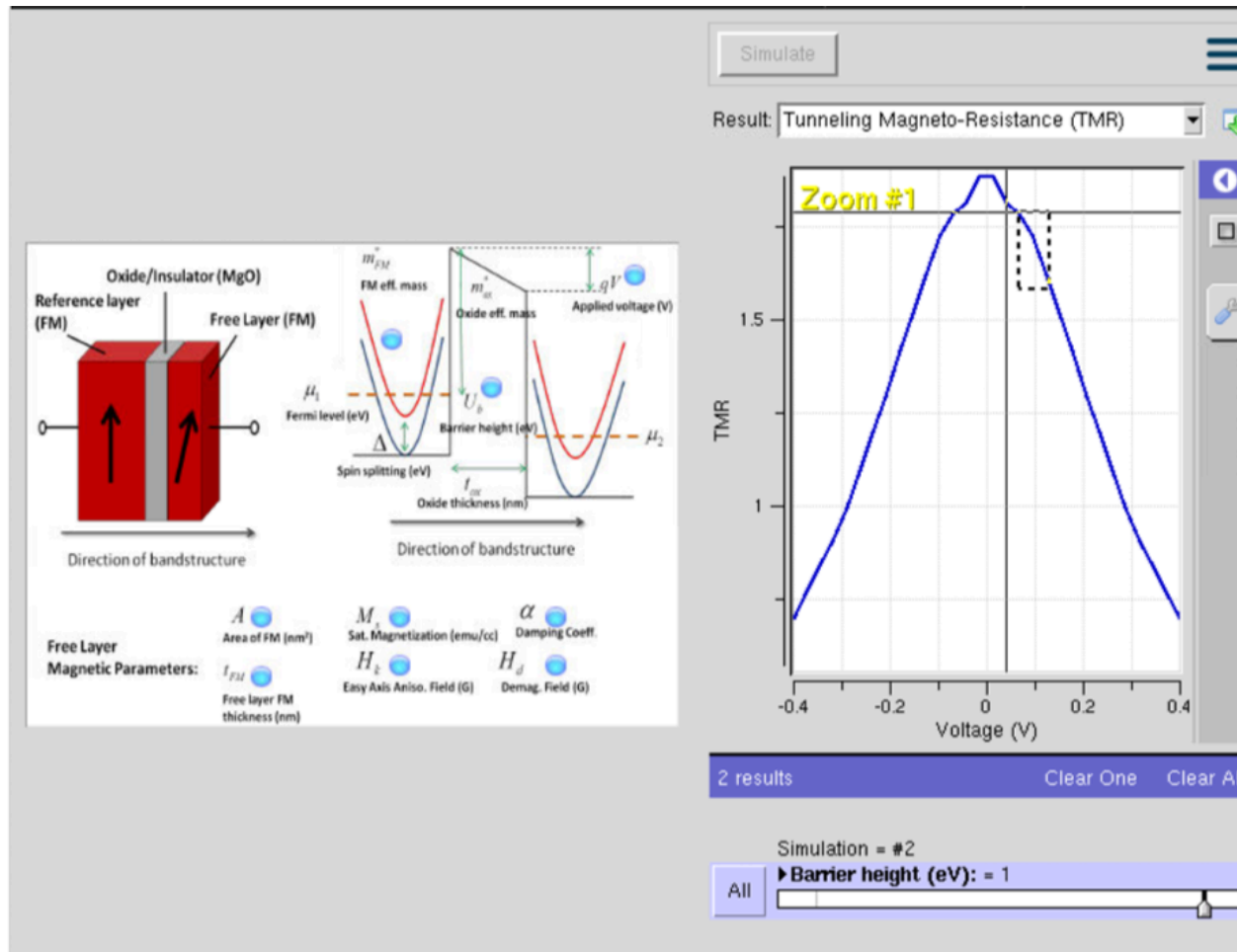
Conclusion: The GFET demonstrated excellent carrier transport and high-speed operation. Device characteristics varied with channel length and gate dielectric material. The simulation illustrates the potential of graphene for future nanoelectronic devices.

Exp. 9. Simulation of Tunneling Magnetoresistance (TMR) by Varying Barrier Height using NanoHUB

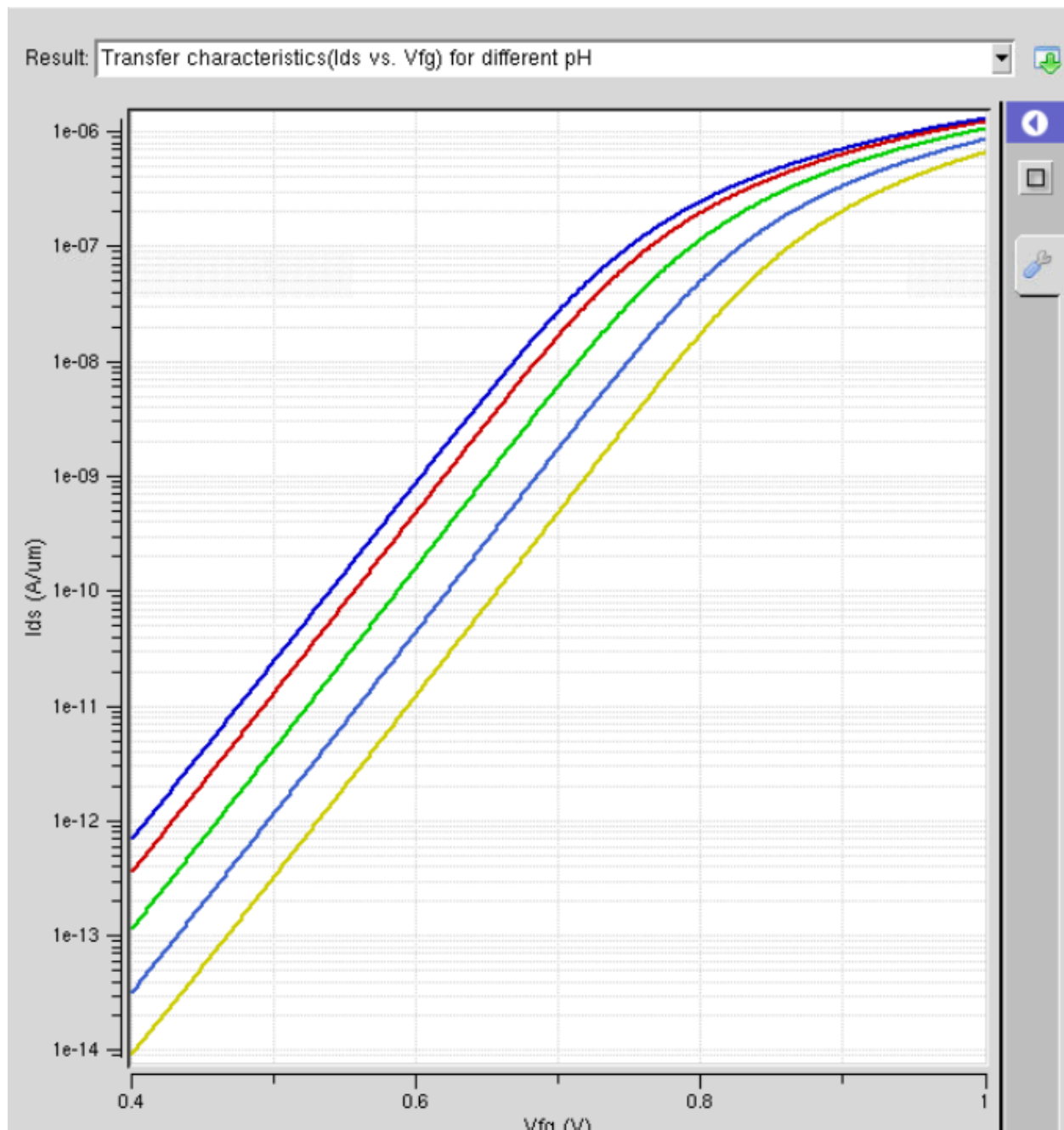
Case1: Characteristics of TMR with barrier height of 1.0eV



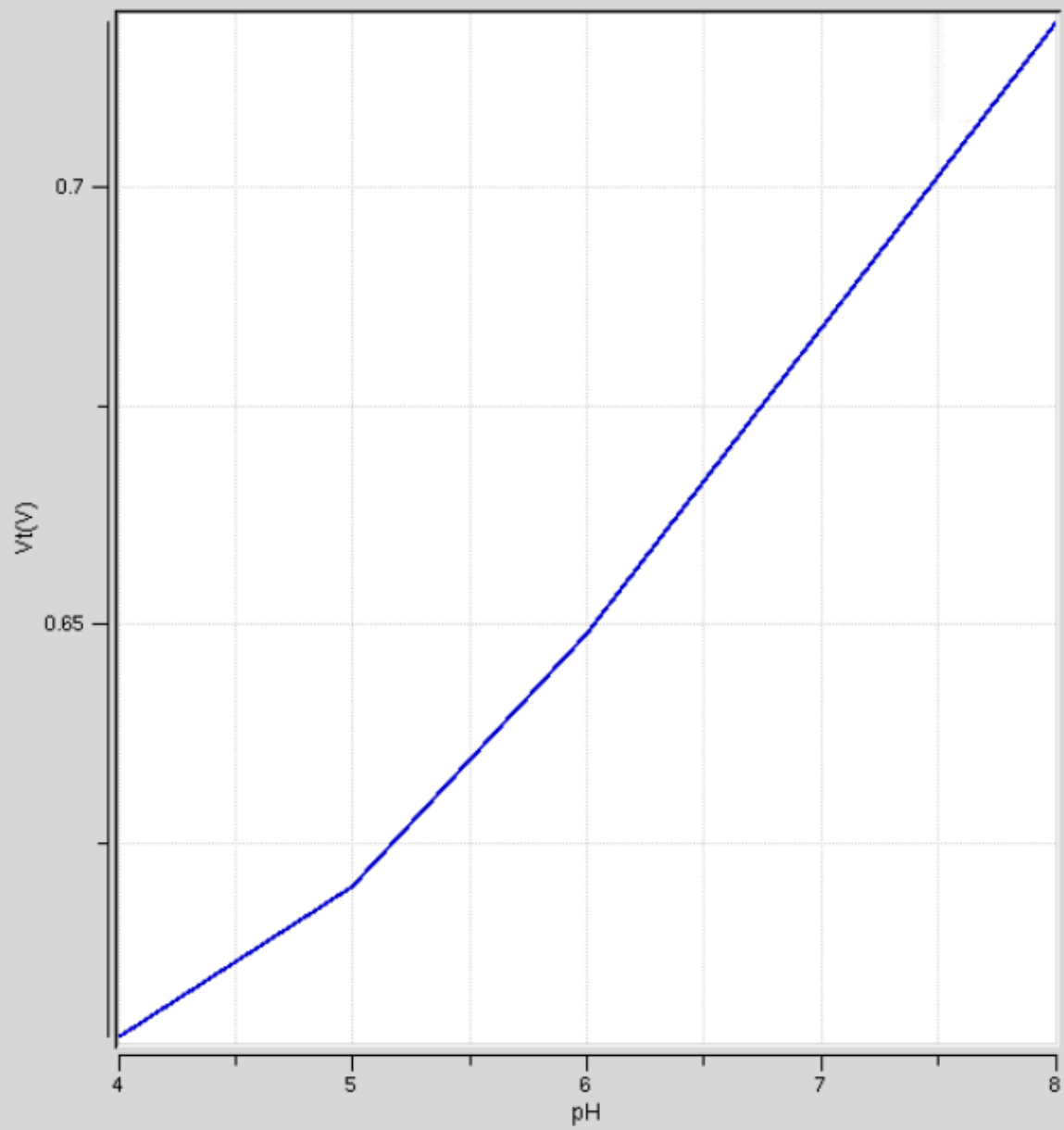
Case2: Characteristics of TMR with barrier height of 0.85eV



Exp. 10. Biosensor and Bio-FET characteristics using NanoHub



Result: Threshold voltage as a function of pH



Exp. 1. Simulation of scattering and absorption of Multi-layer Sphere and Core-Shell Nanoparticle using Mie-Plot tool

Case Study 1: Scattering and absorption spectra for a single-layer sphere

